

ST IGNATIUS COLLEGE RIVERVIEW



ASSESSMENT TASK 4

TRIAL HSC EXAMINATION

YEAR 12

2008

EXTENSION 2

Time allowed: 3 hours (+ 5 minutes reading time)

Instructions to Candidates

- ❖ Attempt all questions.
- ❖ There are eight questions. All questions are of equal value.
- ❖ All necessary working should be shown. Full marks may not be awarded if work is careless or badly arranged.
- ❖ The questions are not necessarily arranged in order of difficulty. Candidates are advised to read the whole paper carefully at the start of the examination.
- ❖ Approved calculators may be used. A table of standard integrals is provided.
- ❖ **Each question is to be started in a new booklet. Your number should be written clearly on the cover of each booklet.**

Question 1 [15 Marks]**Start a new answer booklet.**

(a) Find the following integrals :

(i) $\int \cos^{-1} x \, dx$ [2]

(ii) $\int_0^{\frac{\pi}{2}} \sin^3 x \cos^2 x \, dx$ [2]

(b) (i) Express $\frac{25}{(x+2)(2x-1)^2}$ in partial fractions [3]

(ii) Hence show that $\int_1^2 \frac{25}{(x+2)(2x-1)^2} = \frac{10}{3} - 2 \ln \frac{3}{2}$ [2]

(c) (i) Using the substitution $x = a - t$, or otherwise,

prove that $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$ [2]

(ii) Hence, or otherwise, show that $\int_0^{\frac{\pi}{2}} x \left(\frac{\pi}{2} - x\right) \sin^2 x \, dx = \frac{\pi^3}{96}$ [4]

Question 2 [15 Marks]**Start a new answer booklet.**

(a) If $z = \frac{2-i}{1+i}$, where $z = x + iy$, find $\bar{z}$ in the form $(a + bi)$ [2]

(b) Find the square root of $(21 + 20i)$ in the form $(a + bi)$ [3]

(c) (i) Sketch the locus of $|z + 1 + i| \leq 1$, where $z = x + iy$ [2]

(ii) Find the maximum and minimum values of $|z|$ in part (i) [2]

(d) (i) The complex number $z = x + iy$ is represented by the point P. [3]

If $\frac{z-1}{z-2i}$ is purely imaginary, show that the locus of P is a circle, excluding two points.

(ii) State the centre and the radius of this circle. [1]

(iii) Give the co-ordinates of the two excluded points and the reason for their exclusion. [2]

Question 3 [15 Marks]**Start a new answer booklet.**

- (a) Sketch graphs (on separate number planes) of the following relations, without the use of calculus.

Each graph should be labelled clearly.

(i) $y = (x-1)(x+1)$ [1]

(ii) $y = |x-1|(x+1)$ [2]

(iii) $y = \frac{1}{(x-1)(x+1)}$ [2]

(iv) $y = \sqrt{(x-1)(x+1)}$ [2]

(v) $y = e^{(x-1)(x+1)}$ [2]

(vi) $y = \log_e (x-1)(x+1)$ [2]

- (b) (i) Sketch on the same number plane $y = |x| - 2$ and $y = 4 + 3x - x^2$ [1]

(ii) Hence, or otherwise, solve $\frac{|x|-2}{4+3x-x^2} > 0$ [3]

Question 4 [15 Marks]**Start a new answer booklet.**

- (a) Write down the co-ordinates of the vertices and the foci for the hyperbola $xy = 2$ [3]

- (b) P is a point $(a \sec \theta, b \tan \theta)$ which lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, with centre O .

The tangent P meets the asymptote $y = \frac{b}{a}x$ at Q and the other asymptote at R .

The normal at P meets OQ at K

- (i) Represent the above data with a suitable diagram [1]

- (ii) Derive the equation of the tangent at P [2]

- (iii) Prove that the co-ordinates of Q are [2]

$$(a[\sec \theta + \tan \theta], b[\sec \theta + \tan \theta])$$

- (iv) If the co-ordinates of R are $(a[\sec \theta - \tan \theta], b[\tan \theta - \sec \theta])$ [1]
Prove that P is the midpoint of QR

- (v) (α) If P is equidistant from Q , R and O , prove that the hyperbola is rectangular [2]

- (β) Hence, prove that $Q\hat{K}P = P\hat{O}R$ [4]

Question 5 [15 Marks]**Start a new answer booklet.**

(a) Consider the polynomial $P(x) = x^4 - 2x^3 + 2x - 1$

(i) Show that $P(x) = 0$ has a multiple zero and state its value and multiplicity. [3]

(ii) Hence, fully factorise $P(x)$ [2]

(b) Consider the polynomial $f(x) = ax^3 + bx^2 + cx + d$ where a, b, c and d are real. [5]

Given that two of the roots of $f(x) = 0$ are $(1 - 2i)$ and -2 , and that

$f(-1) = -8$, find a, b, c , and d .

(c) If α, β and γ are the roots of the equation $x^3 + 2x^2 - 3x + 4 = 0$ find [5]

A cubic equation whose roots are $\alpha\beta, \beta\gamma$ and $\gamma\alpha$.

Question 6 [15 Marks]**Start a new answer booklet.**

- (a) A solid has its base the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$ [4]

If each section perpendicular to the major axis is an equilateral triangle,

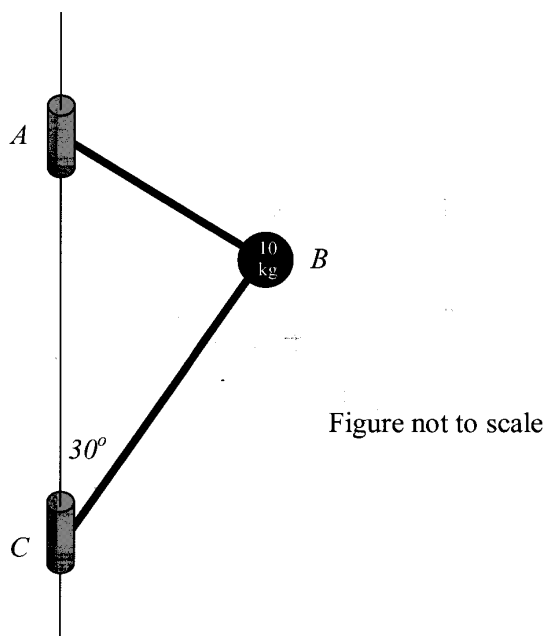
show that the volume of the solid is $128\sqrt{3}$ cubic units.

- (b) The region bounded by the curve $y = \log_e x$, the straight line $x = e$ and the x -axis is rotated about the straight line $x = e$. By taking slices parallel to the x -axis, find the exact volume generated. [5]

- (c) Find the exact volume generated, by rotating the area bound by the curves [6]

$y = (x-1)^2$ and $y = x+1$, about the y -axis using the method of cylindrical shells.

(a)



The above diagram shows a mass of 10 kilograms at B connected by light rods (at right angles) to sleeves A and C which revolve freely about the vertical axis AC but do not move vertically. The angle between the vertical axis AC and the light rod BC is 30° . The acceleration due to gravity is g metres per second squared.

- (i) Given AC is 2 metres, show that the radius of the circular path of rotation of B is $\frac{\sqrt{3}}{2}$ metres. [1]
- (ii) Find the tensions in the rods AB and BC when the mass makes 90 revolutions per minute about the vertical axis. [5]

Question 7 Continued

- (b) A particle P of mass m kg projected vertically upward with an initial velocity u metres per second is subjected to forces which create a constant vertical downward acceleration of magnitude g metres per second squared and an acceleration directed against the motion of magnitude kv when the speed is v metres per second squared. K is a constant

- (i) Show, with the aid of a diagram, that the acceleration function [2]

Is given by $\ddot{x} = -g - kv$

- (ii) Prove that the maximum height reached by the particle after [3]

time T is given by $T = \frac{1}{k} \log_e \left| \frac{g + ku}{g} \right|$

- (ii) Prove that the maximum height is $\frac{1}{k}(u - gT)$ [4]

Question 8 [15 Marks]**Start a new answer booklet.**

- (a) (i) Prove by Mathematical Induction that if n is a positive integer, [4]
then $2^{(n+4)} > (n+4)^2$
- (ii) By choosing a suitable substitution, or otherwise, show that [2]
if a is a positive integer, then $2^{3(a+2)} > 9(a+2)^2$
- (b) (i) Write down the formula for $\tan(A+B)$ in terms of $\tan A$ and $\tan B$ [1]
- (ii) Prove that $\tan(2 \tan^{-1} x) = 2 \tan(\tan^{-1} x + \tan^{-1} x^3)$ [3]
- (c) Consider the curve C in the x - y plane defined by $\sqrt{|x|} + \sqrt{y} = 1$
- (i) Write down the domain for C [1]
- (ii) For $x > 0$, show that $\frac{dy}{dx} < 0$ [2]
- (iii) Sketch a graph of C , paying close attention to the gradient of [2]
the curve at $x = 0$

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad n \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} ax \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Question 1

(a) (i) $I = \int \cos^{-1} x \, dx$

Let $u = \cos^{-1} x \Rightarrow u' = -\frac{1}{\sqrt{1-x^2}}$

$v' = 1 \Rightarrow v = x$

$I = (\cos^{-1} x) \times x - \int \frac{x}{\sqrt{1-x^2}} \, dx$

← This integral can be done by substitution,

$I = x \cos^{-1} x - \sqrt{1-x^2} + C$

(ii) $\int_0^{\pi/2} \sin^3 x \cos^2 x \, dx$

$I = \int_0^{\pi/2} \sin^2 x \sin x \cos^2 x \, dx$

$I = \int_0^{\pi/2} (1 - \cos^2 x) \cos^2 x \sin x \, dx$

Note $\sin^2 x = 1 - \cos^2 x$

$I = -\int_1^0 (1-u^2) u^2 \, du$

Let $u = \cos x$

$\therefore du = -\sin x \, dx$

When $x = \frac{\pi}{2} \Rightarrow u = 0$

$x = 0 \Rightarrow u = 1$

$I = \int_0^1 (u^2 - u^4) \, du$

$= \left[\frac{u^3}{3} - \frac{u^5}{5} \right]_0^1$

$= \left(\frac{1}{3} - \frac{1}{5} \right) - 0$

$= \frac{2}{15}$

Question 1

$$(b) (i) \text{ Let } \frac{25}{(x+2)(2x-1)^2} \equiv \frac{a}{x+2} + \frac{10}{2x-1} + \frac{c}{(2x-1)^2}$$

$$\therefore 25 \equiv a(2x-1)^2 + b(x+2)(2x-1) + c(x+2)$$

$$25 \equiv x^2(4a+2b) + x(3b+c-4a) + (2c-2b-a)$$

Equating co-efficients.

$$4a+2b=0 \Rightarrow 2a+b=0 \quad \text{--- (1)}$$

$$3b+c-4a=0 \quad \text{--- (2)}$$

$$2c-2b+a=25 \quad \text{--- (3)}$$

Solving from (1) $b = -2a$

Sub. in (2) and (3)

$$c-10a=0 \quad \text{--- (2a)}$$

$$2c+5a=25 \quad \text{--- (3a)}$$

$$20a+5a=25$$

$$a=1$$

$$\therefore b=-2$$

$$c=10$$

$$(ii) I = \int_1^2 \frac{25}{(x+2)(2x-1)^2} dx$$

$$= \int_1^2 \left[\frac{1}{x+2} - \frac{2}{2x-1} + \frac{10}{(2x-1)^2} \right] dx$$

$$= \left[\ln|x+2| - \ln|2x-1| - 5(2x-1)^{-1} \right]_1^2$$

$$= \left[\ln 4 - \ln 3 - \frac{5}{3} - (\ln 3 - \ln 1 - 5) \right]$$

$$= \ln 4 - \ln 3 - \frac{5}{3} - \ln 3 + 0 + 5$$

$$= \frac{10}{3} - 2\ln 2 - 2\ln 3 = \frac{10}{3} - 2\ln\left(\frac{3}{2}\right)$$

Question 1

$$\begin{aligned}
 \text{(c) (i)} \quad I &= \int_0^a f(x) dx & \left| \begin{array}{l} \text{let } x = a - t \\ dx = -dt \\ \text{when } x = a, t = 0 \\ x = 0, t = a \end{array} \right. \\
 &= - \int_a^0 f(a-t) dt \\
 &= \int_0^a f(a-t) dt \\
 &= \int_0^a f(a-x) dx & \left| \begin{array}{l} \text{let } x = t \\ \therefore dx = dt \end{array} \right.
 \end{aligned}$$

$$\text{(ii)} \quad I = \int_0^{\frac{\pi}{2}} x \left(\frac{\pi}{2} - x \right) \sin^2 x \, dx \quad \text{--- [A]}$$

$$= \int_0^{\pi/2} \left(\frac{\pi}{2} - x \right) \left[\frac{\pi}{2} - \left(\frac{\pi}{2} - x \right) \right] \sin^2 \left(\frac{\pi}{2} - x \right) dx$$

$$I = \int_0^{\pi/2} \left(\frac{\pi}{2} - x \right) x \cos^2 x \, dx \quad \text{--- [B]}$$

$$2I = A + B$$

$$= \int_0^{\pi/2} x \left(\frac{\pi}{2} - x \right) \sin^2 x \, dx + \int_0^{\pi/2} x \left(\frac{\pi}{2} - x \right) \cos^2 x \, dx$$

$$= \int_0^{\pi/2} x \left(\frac{\pi}{2} - x \right) \left[\sin^2 x + \cos^2 x \right] dx$$

Note $\cos^2 x + \sin^2 x = 1$

$$2I = \int_0^{\pi/2} x \left(\frac{\pi}{2} - x \right) dx = \int_0^{\pi/2} \left(\frac{\pi x}{2} - x^2 \right) dx$$

$$2I = \left[\frac{\pi x^2}{4} - \frac{x^3}{3} \right]_0^{\pi/2} = \frac{\pi^3}{16} - \frac{\pi^3}{24} = \frac{\pi^3}{48}$$

$$I = \frac{\pi^3}{96}$$

Question 2

$$(a) Z = \frac{2-i}{1+i} \times \frac{1-i}{1-i} = \frac{2-3i+i^2}{1-i^2} = \frac{2-3i-1}{1+1} = \frac{1-3i}{2}$$

$$\overline{Z} = \frac{1}{2} + \frac{3}{2}i$$

(b) Let $\sqrt{21+20i} = a+bi$ where a and b are real. #
square both sides

$$21+20i = a^2+2abi+b^2i^2$$

$$21+20i = a^2-b^2+2abi$$

equate real and imaginary parts.

$$a^2-b^2=21 \quad \text{--- (1)}$$

$$ab=10 \quad \text{--- (2)}$$

Sub $b = \frac{10}{a}$ in (1)

$$a^2 - \frac{100}{a^2} = 21$$

$$a^4 - 21a^2 - 100 = 0$$

$$(a^2-25)(a^2+4) = 0$$

$$a^2-25=0 \text{ only \#}$$

$$a = \pm 5$$

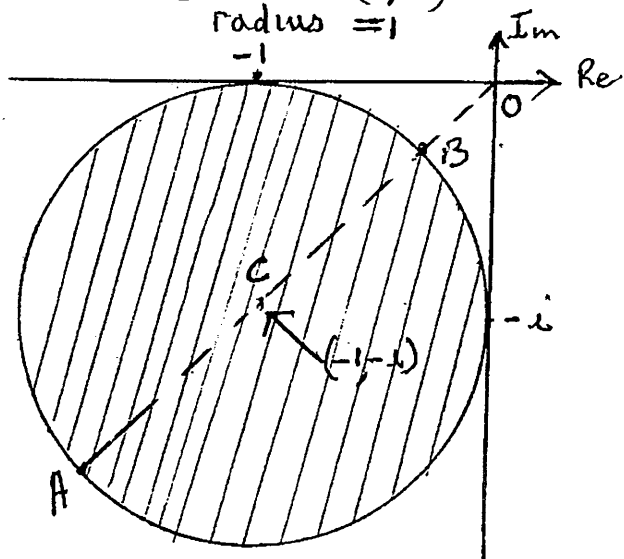
When $a=5$, $b=2$

$a=-5$, $b=-2$

square roots are $(5+2i)$ and $(-5-2i)$

$$(c) |Z - (-1-i)| \leq 1$$

Circle Centre $(-1,-1)$
radius = 1



(ii) Maximum value of $|Z| = OA$

$$OA = OB + AB$$

$$(\sqrt{2}-1) + 2$$

$$= \sqrt{2} + 1$$

Minimum value of $|Z| = OB$

$$OB = OC - BC$$

$$= \sqrt{2} - 1$$

Question 2

(d)(i) If $z = x + iy$

$$\begin{aligned}\frac{z-1}{z-2i} &= \frac{x+iy-1}{x+iy-2i} = \frac{(x-1)+iy}{x+(y-2)i} \times \frac{x-(y-2)i}{x-(y-2)i} \\ &= \frac{x(x-1) - y(y-2)i^2 + (x-1)(y-2)i + xyi}{x^2 - (y-2)^2 i^2} \\ &= \frac{x(x-1) + y(y-2) + (xy - xy + 2x + y - 2)i}{x^2 + (y-2)^2}\end{aligned}$$

But $\frac{z-1}{z-2i}$ is purely imaginary.

Hence $\frac{x(x-1) + y(y-2)}{x^2 + (y-2)^2} = 0$ i.e. the real part is zero.

$$x^2 - x + y^2 - 2y = 0.$$

$$x^2 - x + \frac{1}{4} + y^2 - 2y + 1 = 1 + \frac{1}{4}$$

$$\left(x - \frac{1}{2}\right)^2 + (y-1)^2 = \frac{5}{4} = \left(\frac{\sqrt{5}}{2}\right)^2.$$

(ii) This is a circle with centre $\left(\frac{1}{2}, 1\right)$ and radius $\frac{\sqrt{5}}{2}$ units.

(iii) Now $\left(\frac{z-1}{z-2i}\right)$ is undefined if z is $(0, 2)$ i.e. $(0 + 2i)$

and for $(1, 0)$, $\left(\frac{z-1}{z-2i}\right)$ would be zero $\notin$ Reals,
which does not meet the stated condition.

Note An alternative argument can be used by considering

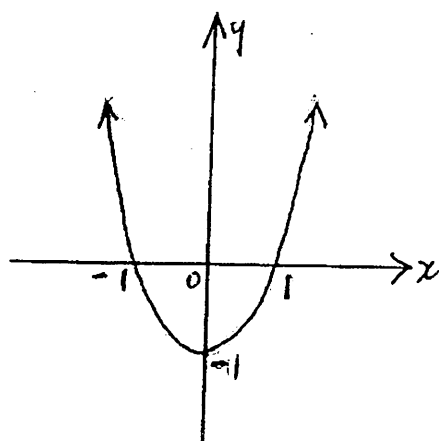
$$\arg\left(\frac{z-1}{z-2i}\right) = \arg(z-1) - \arg(z-2i) = \pm \frac{\pi}{2}$$

The locus is a semicircle above and below.

The join of $P(1, 0)$ and $Q(0, 2)$ excluding P and Q .

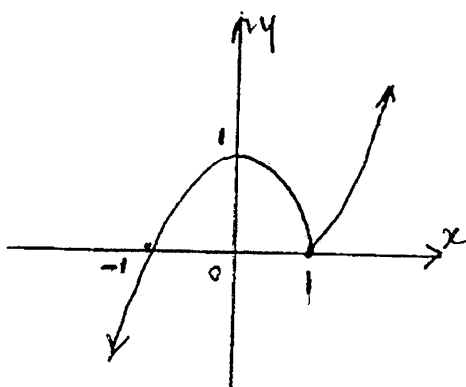
Question 3

(a) (i)



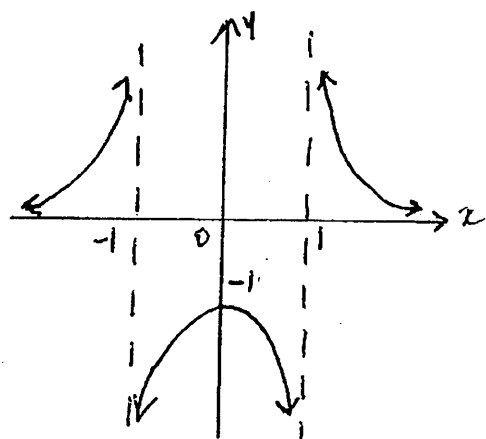
$$y = (x-1)(x+1)$$

(ii)



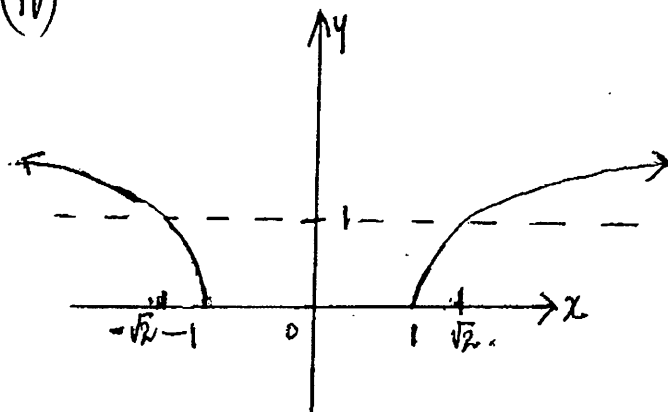
$$y = |x-1|(x+1)$$

(iii)



$$y = \frac{1}{(x-1)(x+1)}$$

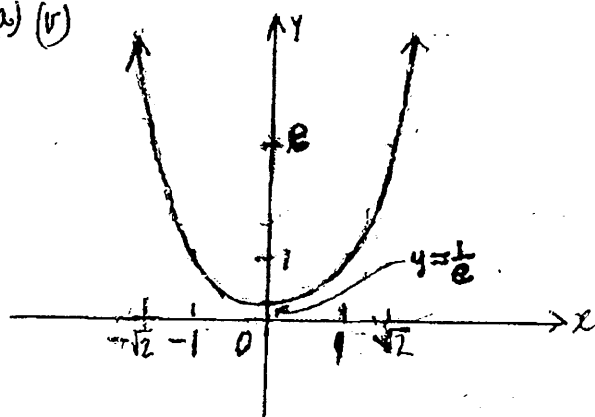
(iv)



$$y = \sqrt{(x-1)(x+1)}$$

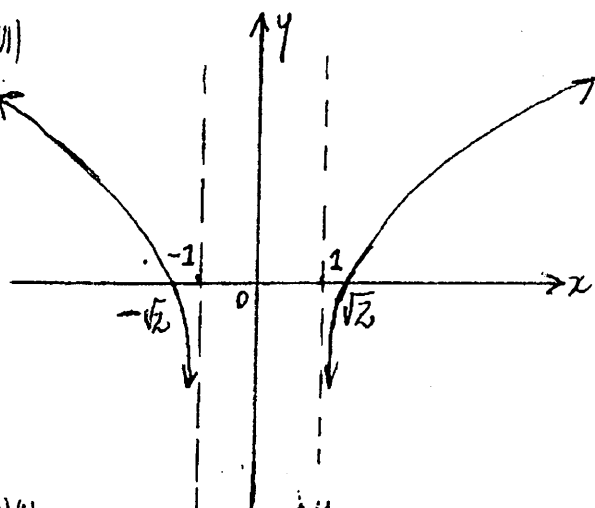
Question 3

(a) (i)



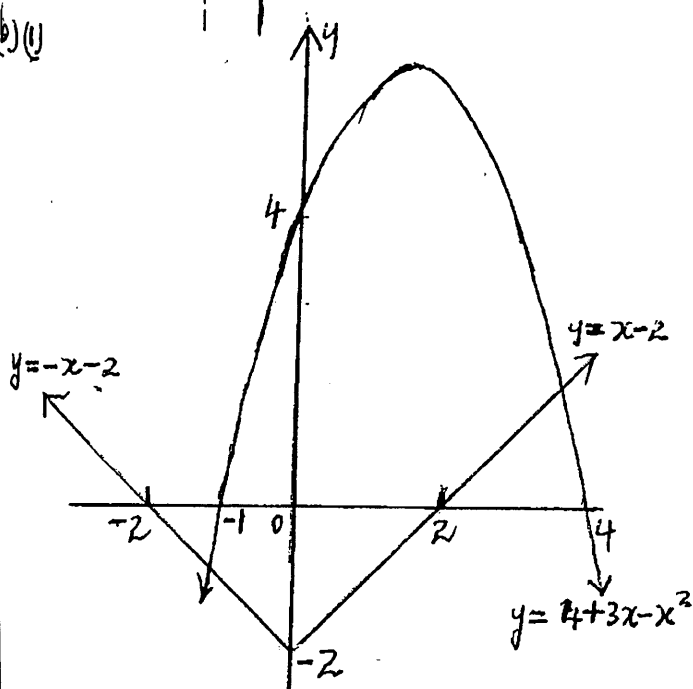
$$y = e^{(x-1)(x+1)}$$

(ii)



$$y = \log_e((x-1)(x+1))$$

(b) (i)



(ii) Now for $\frac{|x|-2}{4+3x-x^2} > 0$

$y = |x| - 2$ and $y = 4 + 3x - x^2$
must have the same sign
Both are positive for $\underline{-2 < x < 4}$
Both are negative for $\underline{-2 < x < -1}$

Hence the solution for the inequality will contain both of these sets.

$$y = 4 + 3x - x^2$$

$$y = -(x^2 - 3x - 4)$$

$$y = -(x-4)(x+1)$$

Question 4

(a) $xy = 2$

Eccentricity $e = \sqrt{2}$

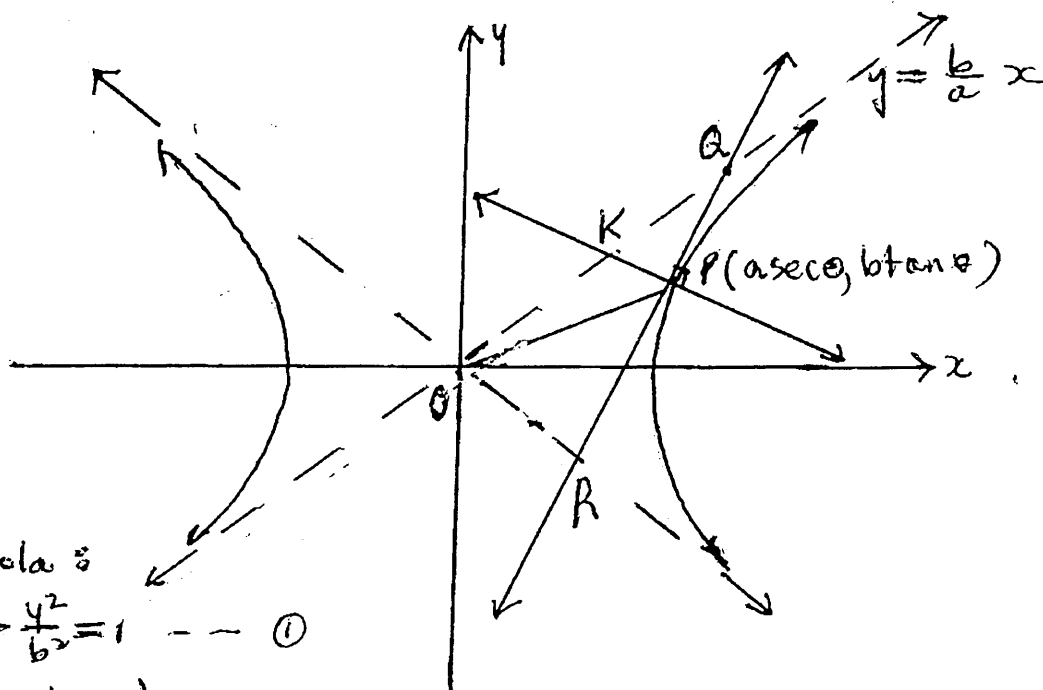
Foci are $(c\sqrt{2}, c\sqrt{2})$ and $(-c\sqrt{2}, -c\sqrt{2})$; Now $c = \sqrt{2}$

$\therefore$ foci are $(2, 2)$ and $(-2, -2)$

Vertices (c, c) and $(-c, -c)$

When $c = 2$ $(\sqrt{2}, \sqrt{2})$ and $(-\sqrt{2}, -\sqrt{2})$

(b) (i)



Hyperbola :

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{--- (1)}$$

$P(a \sec \theta, b \tan \theta)$

Asymptote OK : $y = \frac{b}{a}x$ --- (2)

(ii) tangent at P : $\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$

$$\frac{dy}{dx} = \frac{b^2 x}{a^2 y}$$

Gradient of tangent $m_1 = \frac{b^2 a \sec \theta}{a^2 \times b \tan \theta} = \frac{b \sec \theta}{a \tan \theta}$

Tangent : $y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$

$$ay \tan \theta - ab \tan^2 \theta = bx \sec \theta - ab \sec^2 \theta$$

$$bx \sec \theta - ay \tan \theta = ab \sec^2 \theta - ab \tan^2 \theta$$

Question 4(b)

(i) $b x \sec \theta - a y \tan \theta = ab (\sec^2 \theta - \tan^2 \theta)$

$\therefore ab : \frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1 \dots \textcircled{2}$ Note $\sec^2 \theta - \tan^2 \theta = 1$

(iii) Solving $\textcircled{2}$ and $\textcircled{3}$

Sub $\textcircled{2}$ and $\textcircled{3}$

$$\frac{x \sec \theta}{a} - \frac{b}{a} \left(\frac{\tan \theta}{b} \right) x = 1$$

$$\frac{x \sec \theta}{a} - \frac{x \tan \theta}{a} = 1$$

$$\frac{x}{a} (\sec \theta - \tan \theta) = 1$$

$$x = \frac{a}{\sec \theta - \tan \theta} \times \left(\frac{\sec \theta + \tan \theta}{\sec \theta + \tan \theta} \right)$$

$$x = a (\sec \theta + \tan \theta)$$

Sub in $\textcircled{2}$ $y = \frac{b}{a} \times a (\sec \theta + \tan \theta) = b (\sec \theta + \tan \theta)$

$\therefore Q$ is $(a [\sec \theta + \tan \theta], b [\sec \theta + \tan \theta])$

(iv) For the co-ordinates of the midpoint of QR.

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$M = \left(\frac{a [\sec \theta + \tan \theta] + a (\sec \theta - \tan \theta)}{2}, \frac{b (\sec \theta + \tan \theta) + b (\tan \theta - \sec \theta)}{2} \right)$$

$$M = (a \sec \theta, b \tan \theta)$$

Now M is P .

Question 4

(b) (v) Consider $OP = PQ$

$$ie OP^2 = PQ^2$$

$$\text{using } d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$(a \sec \theta)^2 + (b \tan \theta)^2 = (a \sec \theta - a(\sec \theta + \tan \theta))^2 + (b \tan \theta - b(\sec \theta + \tan \theta))^2$$

$$a^2 \sec^2 \theta + b^2 \tan^2 \theta = a^2 + a^2 \tan^2 \theta + b^2 \sec^2 \theta$$

$$a^2 \sec^2 \theta - a^2 \tan^2 \theta = b^2 \sec^2 \theta - b^2 \tan^2 \theta$$

$$a^2 (\sec^2 \theta - \tan^2 \theta) = b^2 (\sec^2 \theta - \tan^2 \theta)$$

which is $a^2 = b^2$ the condition for a rectangular hyperbola.

(iv) $\widehat{KOR} = 90^\circ$ (rectangular hyperbola $\Rightarrow$ asymptotes $\perp$)
 $\widehat{KPR} = 90^\circ$ (tangent $\perp$ Normal)

$\therefore$ ORPK is a cyclic Quadrilateral.

$\widehat{OKP} = \widehat{PRO}$ (External $\angle$ of a cyclic Quadrilateral equals interior opposite angle)

But $\widehat{PRO} = \widehat{POR}$ (Isosceles ΔPRO , $PR = OR$)

$$\therefore \underline{\underline{\widehat{OKP} = \widehat{POR}}}$$

Question 5.

(a) (i) $P(x) = x^4 - 2x^3 + 2x - 1$

$$P'(x) = 4x^3 - 6x^2 + 2$$

$$P''(x) = 12x^2 - 12x$$

$$= 12x(x-1)$$

When $P''(x) = 0$

$$x = 0 \text{ or } 1$$

Consider $x=1$

$$P'(1) = 4 - 6 + 2 = 0$$

$$P(1) = 1 - 2 + 2 - 1 = 0$$

Now $P(1) = P'(1) = P''(1)$

Hence $x=1$ is a root of multiplicity 3 for $P(x)=0$

(ii) Now $(x-1)^3$ as a factor of $P(x)$

Now consider $x=-1$ in $P(x)$

$$P(-1) = (-1)^4 - 2(-1)^3 + 2(-1) - 1$$

$$= 1 + 2 - 2 - 1 = 0$$

$\therefore (x+1)$ as a factor.

Hence $P(x) = (x-1)^3(x+1)$.

Question 5

(b) $f(x) = ax^3 + bx^2 + cx + d$,

Considering $f(x) = 0$

Now one root is $(1-2i)$ and its conjugate $(1+2i)$ is a root because the co-efficients of $f(x)$ are real.

So the 3 roots of $f(x) = 0$ are $(1 \pm 2i)$ and 2 ,

IF $f(-1) = -8$ Then

$$-a + b - c + d = -8$$

$$\text{ie } a - b + c - d = 8 \quad \text{--- --- (1)}$$

Sum of roots : $\alpha + \beta + \gamma = -\frac{b}{a}$

$$1+2i + 1-2i + 2 = -\frac{b}{a}$$

$$0 = -\frac{b}{a}$$

$$\boxed{b = 0} \quad \text{--- --- (2)}$$

Product of roots : $\alpha\beta\gamma = -\frac{d}{a}$

$$(1-2i)(1+2i)(-2) = -\frac{d}{a}$$

$$-2(1-2i^2) = \frac{d}{a}$$

$$d = 10a \quad \text{--- --- --- (3)}$$

Sum of roots 2 at a time : $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$

$$(1+2i)(1-2i) + (1+2i)(-2) + (1-2i)(-2) = \frac{c}{a}$$

$$5 + -2 + 4i - 2 + 4i = \frac{c}{a}$$

$$1 = \frac{c}{a}$$

$$c = a \quad \text{--- --- (4)}$$

sub (2), (3), (4) in (1) $a + 0 + a - 10a = 8$

$$\underline{\underline{a = -1, c = -1, d = -10.}}$$

Question 5

(c) $x^3 + 2x^2 - 3x + 4 = 0$ ----- [A]

roots α, β, γ

$\alpha + \beta + \gamma = -2$ ----- ①

$\alpha\beta + \alpha\gamma + \beta\gamma = -3$ ----- ②

$\alpha\beta\gamma = -4$ ----- ③

Now $\alpha\beta + \beta\gamma + \gamma\alpha =$ sum for required equation.

$$\begin{aligned} \alpha\beta \times \beta\gamma + \alpha\beta \times \gamma\alpha + \beta\gamma \times \gamma\alpha &= \alpha\gamma\beta^2 + \beta\gamma\alpha^2 + \alpha\beta\gamma^2 \\ &= \alpha\beta\gamma(\alpha + \beta + \gamma) = -4 \times -2 = 8 \end{aligned}$$

This is the ^{of roots} sum, taken 2 at a time for the required equation.

$\alpha\beta \times \beta\gamma \times \gamma\alpha = \alpha^2\beta^2\gamma^2 = (\alpha\beta\gamma)^2 = (-4)^2 = 16$.

This is the product of the roots for the required equation.

Required equation:

$$x^3 - \left(\begin{matrix} \text{Sum of} \\ \text{roots} \end{matrix} \right) x^2 + \left(\begin{matrix} \text{sum of} \\ \text{roots 2 at} \\ \text{a time} \end{matrix} \right) x - \left(\begin{matrix} \text{product of} \\ \text{roots} \end{matrix} \right) = 0$$

i.e. $x^3 + 3x^2 + 8x - 16 = 0$

Note A more elegant approach, try: $\alpha\beta\gamma(\frac{1}{\gamma}); \alpha\beta\gamma(\frac{1}{\beta}); \alpha\beta\gamma(\frac{1}{\alpha})$

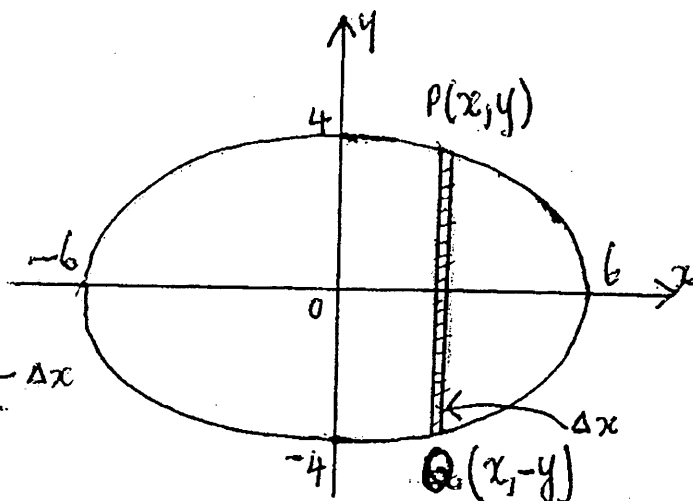
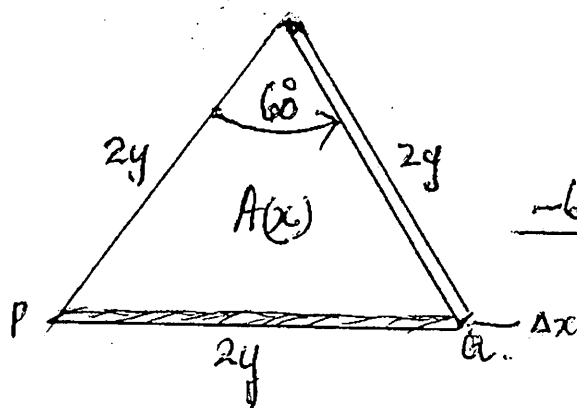
$\Rightarrow -\frac{4}{\gamma}; -\frac{4}{\beta}; -\frac{4}{\alpha}$

and use the transformation: $y = -\frac{4}{x}$

Hence sub $x = -\frac{4}{y}$ in [A] above etc. B

Question 6

(a) $\frac{x^2}{36} + \frac{y^2}{16} = 1$



$$A(x) = \frac{1}{2} \times 2y \times 2y \times \sin 60^\circ$$

$$= 2y^2 \times \frac{\sqrt{3}}{2}$$

$$= y^2 \sqrt{3}$$

$$A(x) = 16 \left(1 - \frac{x^2}{36}\right) \sqrt{3}$$

$$\Delta V \approx 16\sqrt{3} \left(1 - \frac{x^2}{36}\right) \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=-6}^{x=6} 16\sqrt{3} \left(1 - \frac{x^2}{36}\right) \Delta x$$

$$V = 16\sqrt{3} \int_{-6}^6 \left(1 - \frac{x^2}{36}\right) dx$$

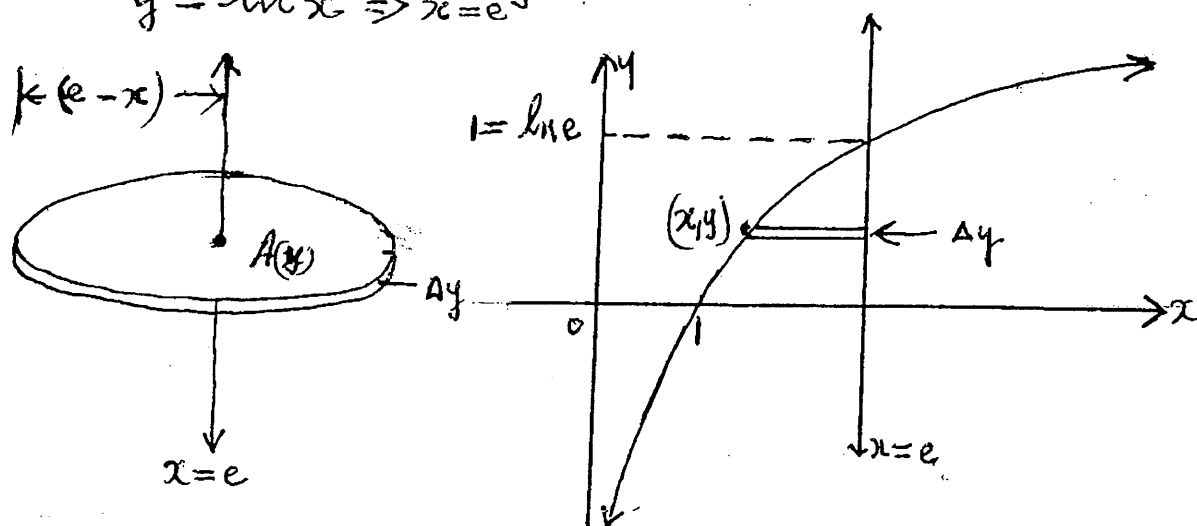
$$V = 32\sqrt{3} \int_0^6 \left(1 - \frac{x^2}{36}\right) dx$$

$$= 32\sqrt{3} \left[x - \frac{x^3}{108} \right]_0^6 = 32\sqrt{3} \left[6 - \frac{216}{108} \right] = 32\sqrt{3} \times 4$$

$$= \underline{\underline{128\sqrt{3} \text{ units}^3}}$$

Question 6

(b) $y = \ln x \Rightarrow x = e^y$



$$A(y) = \pi(e-x)^2$$

$$\Delta V \doteq \pi(e-x)^2 \Delta y$$

$$\Delta V \doteq \pi(e - e^y)^2 \Delta y$$

$$V = \lim_{\Delta y \rightarrow 0} \sum_{y=0}^{y=1} \pi(e^2 - 2e y^e + e^{2y}) \Delta y$$

$$V = \pi \int_0^1 (e^2 - (2e)y^e + e^{2y}) dy$$

$$V = \pi \left[e^2 y - (2e)y^e + \frac{1}{2} e^{2y} \right]_0^1$$

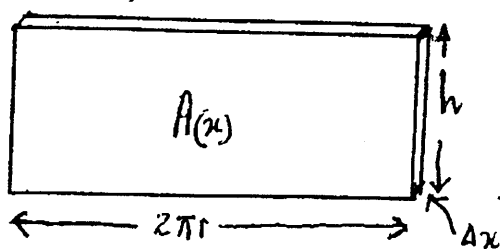
$$V = \pi \left[\left(e^2 - 2e^2 + \frac{e^2}{2} \right) - \left(0 - 2e + \frac{1}{2} \right) \right]$$

$$V = \pi \left(-\frac{e^2}{2} + 2e - \frac{1}{2} \right)$$

$$V = \frac{\pi}{2} (4e - e^2 - 1) \text{ units}^3.$$

Question 6

(c) $y = x+1$ --- ①
 $y = (x-1)^2$ --- ②



$$r = x$$

$$h = (x+1) - (x-1)^2$$

$$= 3x - x^2$$

$$A(x) = 2\pi r h$$

$$= 2\pi x(3x - x^2)$$

$$\Delta V \doteq 2\pi(3x^2 - x^3)\Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^{x=3} 2\pi(3x^2 - x^3)\Delta x$$

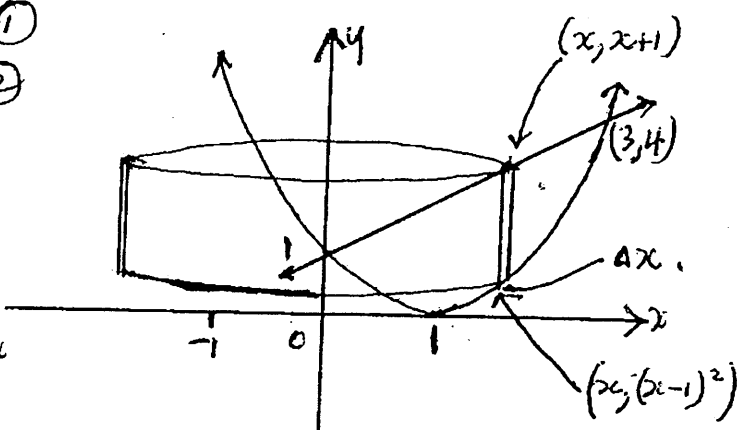
$$V = 2\pi \int_0^3 (3x^2 - x^3) dx$$

$$V = 2\pi \left[x^3 - \frac{x^4}{4} \right]_0^3$$

$$V = 2\pi \left(27 - \frac{81}{4} \right)$$

$$V = 2\pi \left(\frac{108 - 81}{4} \right)$$

$$V = \frac{27\pi}{4} \text{ units}^3$$



Note Intersection of st line and parabola can be obtained by solving simultaneously.

(II)

Re
Vector

Horiz

T₂

T₂

T₂

Solving
from ①

T₂

T₂

4T₂

Question 7

(a) (i) In $\triangle ABC$

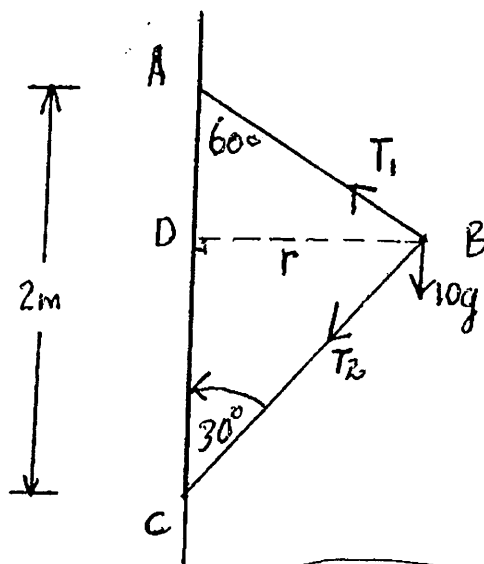
$$\frac{AB}{2} = \sin 30^\circ$$

$$AB = 2 \times \frac{1}{2} = 1 \text{ unit.}$$

In $\triangle ABD$

$$\frac{r}{1} = \sin 60^\circ$$

$$r = \frac{\sqrt{3}}{2} \quad (\text{ie } BD)$$



(ii) Now $2\pi \text{ rad} = 1 \text{ revolution}$

$$\omega = 90 \text{ rpm}$$

$$1 \text{ rev/min} = \frac{2\pi}{60} \text{ rad/sec}$$

$$90 \text{ rpm} = \frac{2\pi}{60} \times 90 \text{ rad/sec}$$

$$= 3\pi \text{ rad/sec}$$

Note for (ii) below, if using $g=10$ Then

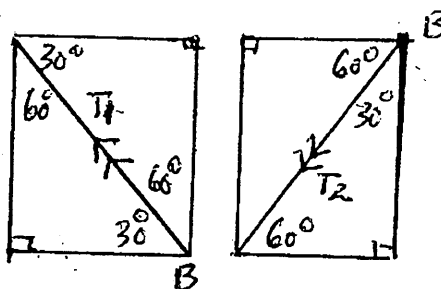
$$\left. \begin{aligned} T_2 &\doteq 290.03 \text{ N} \\ T_1 &\doteq 716.20 \text{ N} \end{aligned} \right\} \text{c.2.d.f.}$$

Resolving Forces at B.

Vertically $T_1 \cos 60^\circ - T_2 \cos 30^\circ = 10g$

$$T_1 \times \frac{1}{2} - T_2 \times \frac{\sqrt{3}}{2} = 10g$$

$$T_1 - T_2\sqrt{3} = 20g \quad \text{--- (1)}$$



Horizontally

$$T_2 \cos 60^\circ + T_1 \cos 30^\circ = m r \omega^2$$

$$T_2 \times \frac{1}{2} + T_1 \times \frac{\sqrt{3}}{2} = 10 \times \frac{\sqrt{3}}{2} \times (3\pi)^2$$

$$T_2 + T_1\sqrt{3} = 90\sqrt{3}\pi^2 \quad \text{--- (2)}$$

Solving (1) and (2)

from (1) $T_1 = 20g + T_2\sqrt{3}$ sub in (2)

$$T_2 + \sqrt{3}(20g + T_2\sqrt{3}) = 90\sqrt{3}\pi^2$$

$$T_2 + 20\sqrt{3}g + 3T_2 = 90\sqrt{3}\pi^2$$

$$4T_2 = 90\sqrt{3}\pi^2 - 20\sqrt{3}g$$

$$T_2 = \frac{90\sqrt{3}\pi^2 - 20\sqrt{3}g}{4}$$

$$T_2 = \frac{5\sqrt{3}(9\pi^2 - 2g)}{2}$$

$$T_1 = \frac{5(27\pi^2 + 2g)}{2}$$

Tension in BC

Tension in AB

Question 7

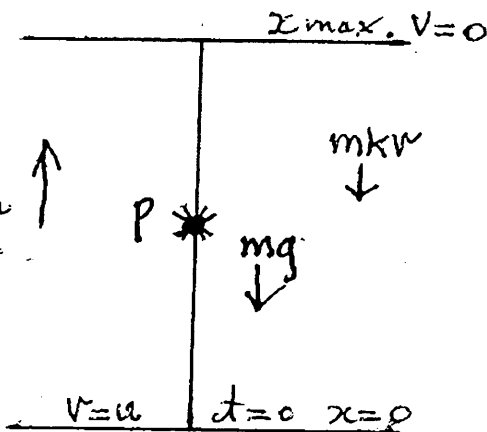
(b) (i)

$$F = ma$$

$$ma = -mg - mkv$$

$$a = -g - kv \quad \text{--- (1)}$$

Direction
+ive $\uparrow$



(ii) let $a = \frac{dv}{dt}$

$$\frac{dv}{dt} = -g - kv$$

$$\frac{dv}{dt} = -(g + kv)$$

$$\frac{dt}{dv} = -\frac{1}{g + kv}$$

$$t = -\frac{1}{k} \ln|g + kv| + c$$

Now when $t = 0$, $v = u$

$$0 = -\frac{1}{k} \ln|g + ku| + c$$

$$c = \frac{1}{k} \ln|g + ku|$$

$$t = \frac{1}{k} \ln|g + ku| - \frac{1}{k} \ln|g + kv|$$

$$t = \frac{1}{k} \ln \left| \frac{g + ku}{g + kv} \right|$$

Now for max height $v = 0$, $t = T$

$$T = \frac{1}{k} \ln \left| \frac{g + ku}{g} \right| \quad \text{--- (2)}$$

Question 7

(b)(ii) back to (b)(i) let $a = v \frac{dv}{dx}$

$$v \frac{dv}{dx} = -(g + kv)$$

$$\frac{dv}{dx} = -\frac{g + kv}{v}$$

$$\frac{dx}{dv} = -\frac{v}{g + kv}$$

$$\frac{dx}{dv} = \frac{g}{k} \left(\frac{1}{g + kv} \right) - \frac{1}{k}$$

Note $\frac{v}{g + kv} = \frac{\frac{1}{k}(g + kv) - \frac{g}{k}}{g + kv}$

or

$$\frac{1}{kv + g} \left/ \begin{array}{l} v + 0 \\ v + \frac{g}{k} \end{array} \right/$$

$$-\frac{g}{k}$$

$$x = \frac{g}{k} \times \frac{1}{k} \ln|g + kv| - \frac{v}{k} + c$$

$$x = \frac{g}{k^2} \ln|g + kv| - \frac{v}{k} + c$$

when $x = 0$, $v = u$.

$$0 = \frac{g}{k^2} \ln|g + ku| - \frac{u}{k} + c$$

$$c = \frac{u}{k} - \frac{g}{k^2} \ln|g + ku|$$

$$\therefore x = \frac{g}{k^2} \ln \left| \frac{g + kv}{g + ku} \right| + \left(\frac{u}{k} - \frac{v}{k} \right)$$

When $v = 0$ for max height

$$x = \frac{g}{k^2} \ln \left| \frac{g}{g + ku} \right| + \frac{u}{k}$$

$$x = -\frac{g}{k^2} \ln \left| \frac{g + ku}{g} \right| + \frac{u}{k}$$

for max height

$$x = -\frac{g}{k^2} T + \frac{u}{k} = \frac{1}{k} (u - gT)$$

using ② from part (b)

Question 8

(a)(i) $2^{(n+4)} > (n+4)^2$

For $n=1$

$$2^5 > 5^2$$

$32 > 25$ which is true

consider the result to be true for $n=k$.

$$\therefore 2^{(k+4)} > (k+4)^2$$

consider $n=k+1$

$$\begin{aligned} 2^{(k+1+4)} &= 2(2^{k+4}) \\ &> 2(k+4)^2 \\ &= 2(k^2 + 8k + 16) \\ &= 2k^2 + 16k + 32 \\ &= k^2 + 10k + 25 + k^2 + 6k + 7 \\ &= (k+5)^2 + k^2 + 6k + 7 \end{aligned}$$

Now since k is a positive integer $k^2 + 6k + 7 > 0$

$$\text{Hence } 2^{(k+5)} > (k+5)^2$$

$$\text{i.e. } 2^{[(k+1)+4]} > [(k+1)+4]^2$$

This is the required inequality i.e. $(k+1)$ in place of n .
So if the result is true for $n=k$, Then it is true for $n=k+1$.

Now the result is true for $n=1$, hence true for $n=2$ and $n=3$ and so on for all positive integers n

Question 8

(a) (i) Now let $3(a+2) = n+4$

$$\therefore 3a+2 = n$$

Replace 'n' by $3a+2$ in the part (i) result.

$$\begin{aligned} 2^{3a+2+4} &> (3a+2+4)^2 \\ &= (3a+6)^2 \\ &= [3(a+2)]^2 \\ &= 9(a+2)^2 \end{aligned}$$

$$\therefore 3^{3(a+2)} > 9(a+2)^2$$

(b) (i) $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

(ii) $\tan(2 \tan^{-1} x) = 2 \tan(\tan^{-1} x + \tan^{-1} x^3)$
Consider RHS = $2 \tan(A+B)$

Where $A = \tan^{-1} x \Rightarrow \tan A = x$

$B = \tan^{-1} x^3 \Rightarrow \tan B = x^3$

$$\text{RHS} = 2 \left(\frac{\tan A + \tan B}{1 - \tan A \tan B} \right)$$

$$= 2 \left(\frac{x + x^3}{1 - x^4} \right)$$

$$= \frac{2x(1+x^2)}{(1-x^2)(1+x^2)}$$

$$= \frac{2x}{1-x^2}$$

$$= \frac{2 \tan A}{1 - \tan^2 A}$$

$$= \tan 2A$$

$$= \tan(2 \tan^{-1} x)$$

$$= \text{LHS}$$

Question 8

(c) $\sqrt{|x|} + \sqrt{y} = 1 \Rightarrow \sqrt{y} = 1 - \sqrt{|x|}$ ($\sqrt{y}$ cannot be less than zero)

(i) $0 \leq |x| \leq 1$

$|x| \leq 1$

$-1 \leq x \leq 1$

(ii) If $x > 0$ Then

$\sqrt{x} + \sqrt{y} = 1$

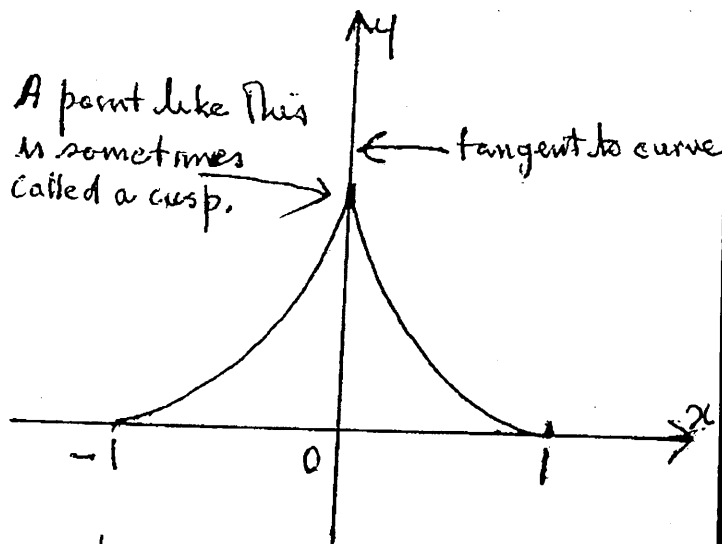
$x^{\frac{1}{2}} + y^{\frac{1}{2}} = 1$

$\frac{1}{2}x^{-\frac{1}{2}} + \frac{1}{2}y^{-\frac{1}{2}} \frac{dy}{dx} = 0$

$\therefore \frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}} < 0$

as both $\sqrt{y}$ and $\sqrt{x}$
are both > 0

(iii) If (x, y) is on curve Then
so is $(-x, y)$ hence the
y axis is an axis of
symmetry



Note when $x=0$, $\frac{dy}{dx}$ is
undefined $\Rightarrow$ vertical
tangent.

Note when $x=\pm 1$, $y=0$

Note $0 \leq y \leq 1$